

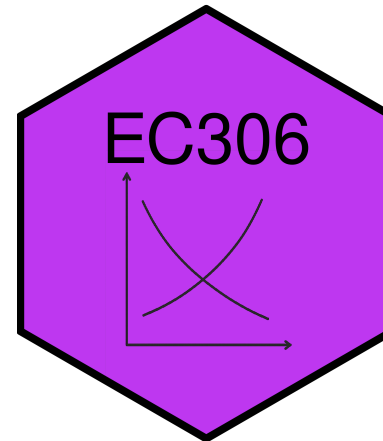
# Labour Supply 2: Labour Supply and Public Policy

EC306 - Wages and Employment

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# Goals of This Section

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Today we will:



# Income Maintenance Programs

# Overview of Income Maintenance Programs

# Government Transfers in Canada

In 2017, 83% of families received some form of government transfer:

| Program                | % Receiving | Average Amount |
|------------------------|-------------|----------------|
| Federal Child Benefits | 20.2%       | \$6,314        |
| Employment Insurance   | 14.0%       | \$8,352        |
| Social Assistance      | 11.3%       | \$8,367        |
| Old Age Security       | 25.1%       | \$8,566        |
| CPP/QPP                | 32.4%       | \$10,092       |

# Issues in Program Design

## Universal vs. Targeted Programs

- Universal: easy to implement but expensive
- Targeted: benefits only those in need but costly to administer

## Work Disincentives

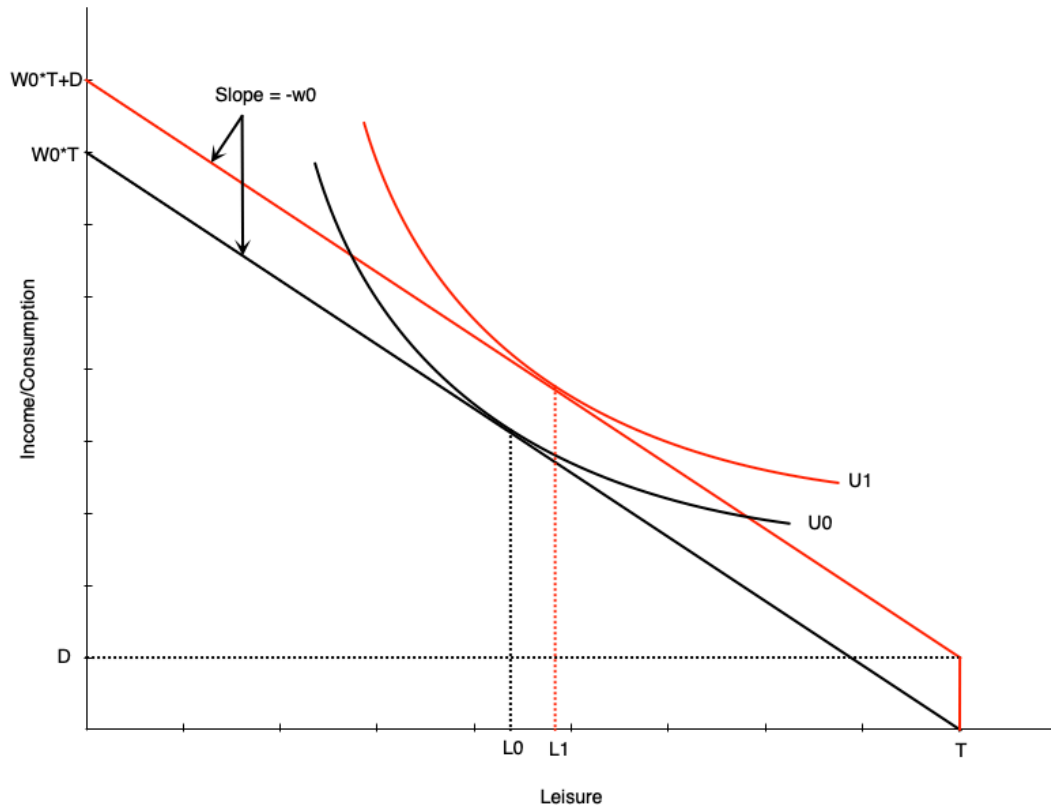
- Some programs reduce incentive to work
- Example: welfare benefits can encourage labour force dropout for those with strong preference for leisure

# Demogrants, Welfare, Wage Subsidies

# Four Standard Income Support Programs

We will examine four policy approaches:

# Demogrant: Theory



## How it works:

- Unconditional lump sum transfer  $D$
- Shifts budget line up parallel
- Wage unchanged → **pure income effect**

## Labor supply impact:

- If leisure is normal good → optimal leisure rises
- **Unambiguously negative work incentive**
- Reduces hours worked
- May induce non-participation

# Demogrant: Canadian Example

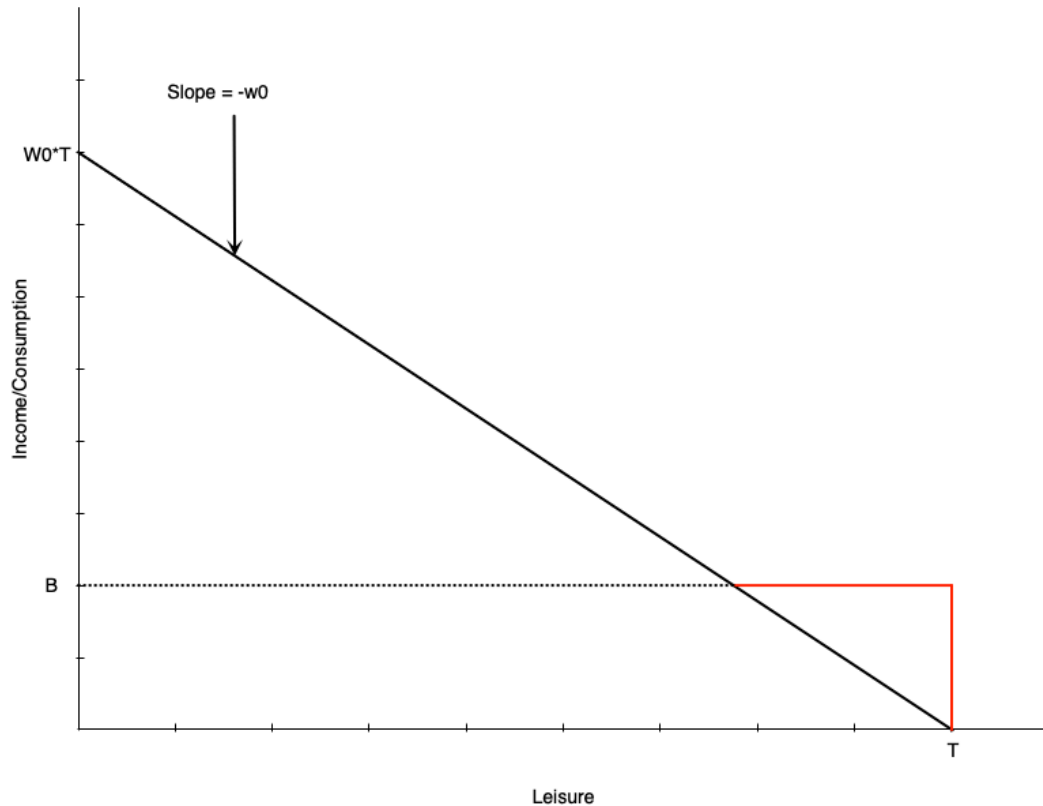
Universal Child Care Benefit (UCCB)

# Welfare: Overview

Welfare = Social/Income Assistance in Canada

- Lump-sum payment to non-workers
- Reduced **dollar-for-dollar** with earned income

# Welfare: Budget Constraint



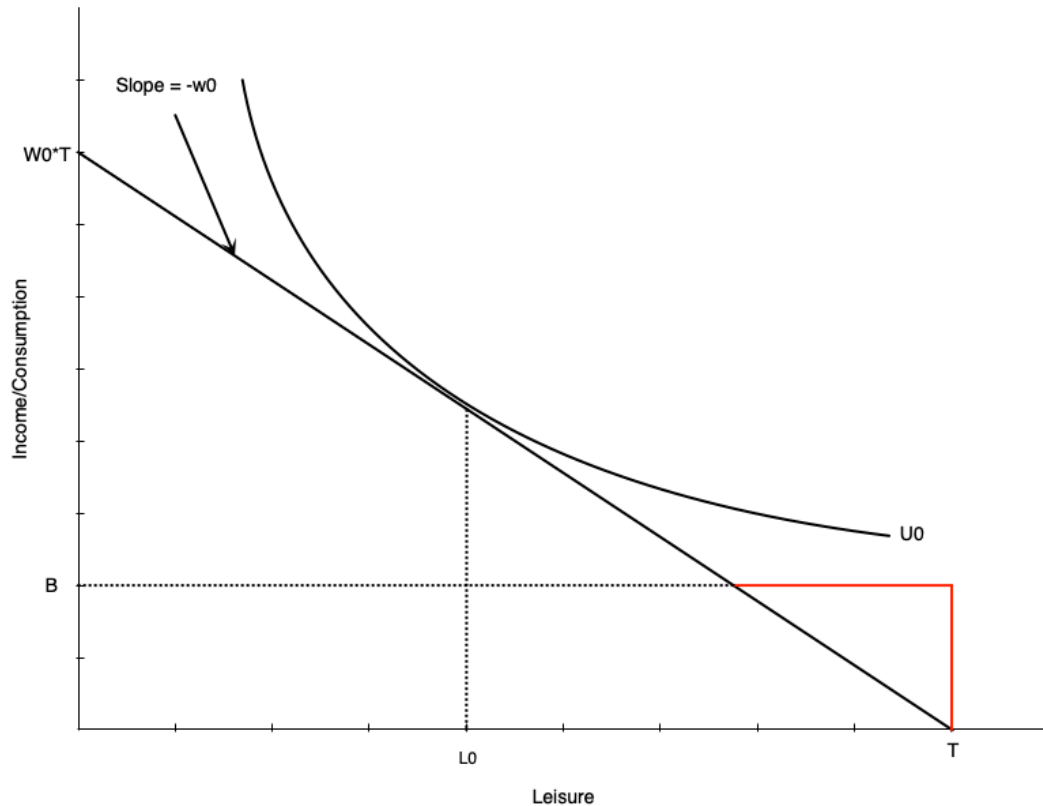
Budget line structure:

- **Black line:** original budget (leisure at  $L_0$ )
- **Red then black:** with welfare benefit  $B$

Key features:

- Benefit  $B$  at zero hours of work
- Reduced \$1 for each \$1 earned
  - Equivalent to **100% tax rate** on earnings
- Income constant at  $B$  until earned income equals benefit
- After that, earn wage  $w_0$  per hour worked

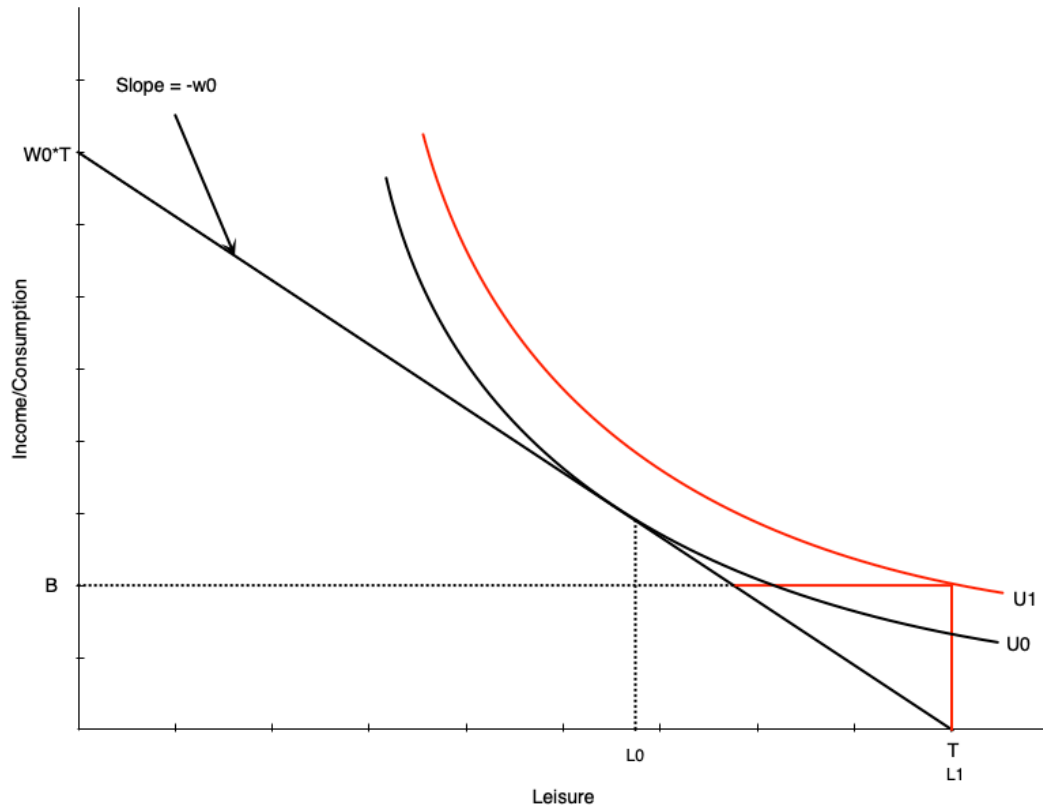
# Welfare: No Effect Case



For some workers, welfare has no effect:

- Person working  $H_0$  hours before benefit
- Still maximizes utility at  $H_0$  hours after
- Generally occurs with people already working many hours

# Welfare: Work Disincentive Case



Others face strong work disincentive:

- Person chooses **zero hours** after welfare benefit

This occurs when:

- Welfare benefit is generous
- Individual places high value on leisure

This is the problematic effect of welfare programs

# Combating Work Disincentives

## Policy options:

### 1. Reduce benefit level

- But may not provide adequate support

### 2. Increase wages

- Through minimum wages, subsidies, training, unions
- Can be expensive and may reduce labour demand

### 3. Change preferences

- Alter attitudes toward work/leisure
- Difficult to achieve

### 4. Negative income tax

- Reduce tax-back rate on earnings
- Discussed next...

# Welfare: Evidence from Quebec

Natural experiment: Welfare benefit structure changed in 1989

## Before 1989:

- Age-based benefits for singles/couples with no kids
- Under 30: \$185/month
- 30 and over: \$507/month

## Results at age 30:

- Much longer time on welfare
- Lower employment rates

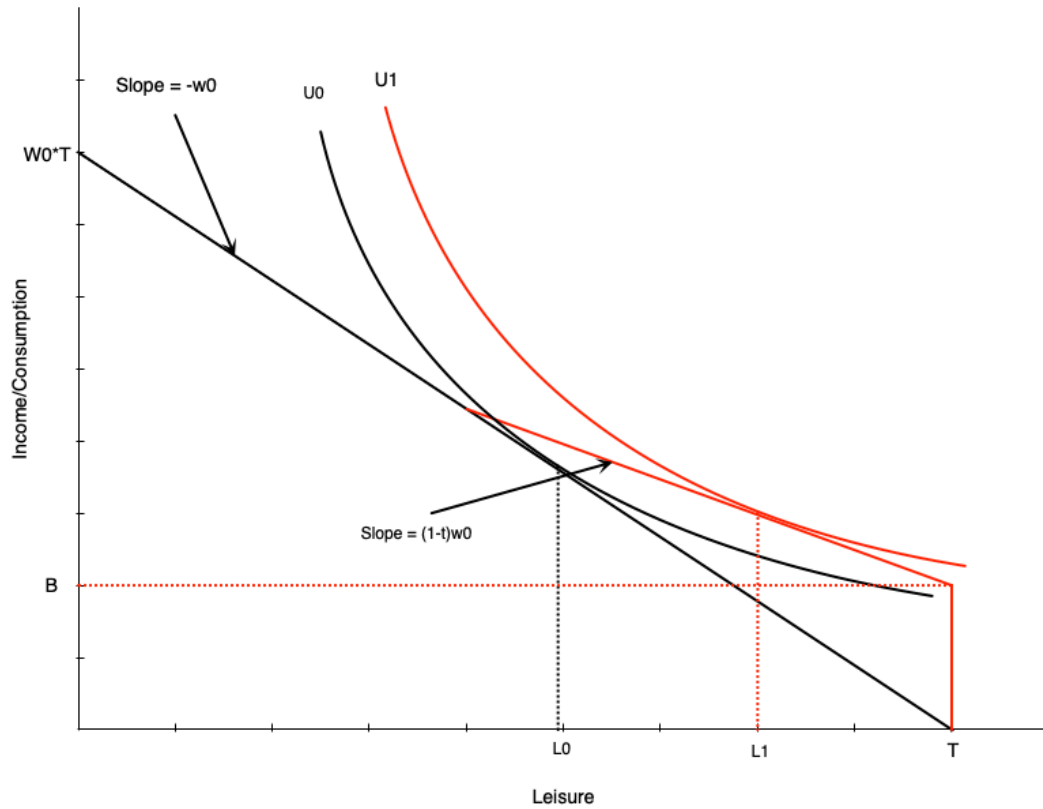
Turning 30 created a **large discontinuous jump** in benefits

# Negative Income Tax: Overview

What is NIT?

- Welfare program where benefit reduced at rate  $< 100\%$  of earnings
- Example: \$10,000 benefit reduced by 50¢ per dollar earned

# Negative Income Tax: Budget Constraint



NIT has two components:

- Benefit  $B$  at zero hours of work
- Tax rate  $t$  on earned income (until earned income equals benefit)

Effect on labour supply:

- Pivots budget line up vs. pure welfare
- Work hours reduced from  $H_0$  to  $H_1$
- **Work disincentive less than pure welfare**
- Person still works some hours (vs. dropping out with pure welfare)

# Negative Income Tax: Income and Substitution Effects

Income effects:

# Negative Income Tax: Math

The budget constraint with NIT:

$$C = \begin{cases} wh + B - twh, & \text{if } B - twh \geq 0 \\ wh, & \text{otherwise} \end{cases}$$

# MINCOME: Manitoba's NIT Experiment

Program details (1970s):

# MINCOME Treatment Groups

|                        |        | Tax Rate (on total Income) |        |        |
|------------------------|--------|----------------------------|--------|--------|
|                        |        | 35%                        | 50%    | 75%    |
| Guarantee at enrolment | \$3800 | Plan 1                     | Plan 3 | *      |
|                        | \$4800 | Plan 2                     | Plan 4 | Plan 7 |
|                        | \$5480 | X                          | Plan 5 | Plan 8 |
|                        |        | Plan 9                     |        |        |

\* Plan 6 was collapsed into Plan 7 because participants viewed this option as too punitive and left the experiment.  
X This option was deemed to expensive for the budget.  
Plan 9 was the control group

Source: Wikipedia



# MINCOME: Results

## Labour supply effects:

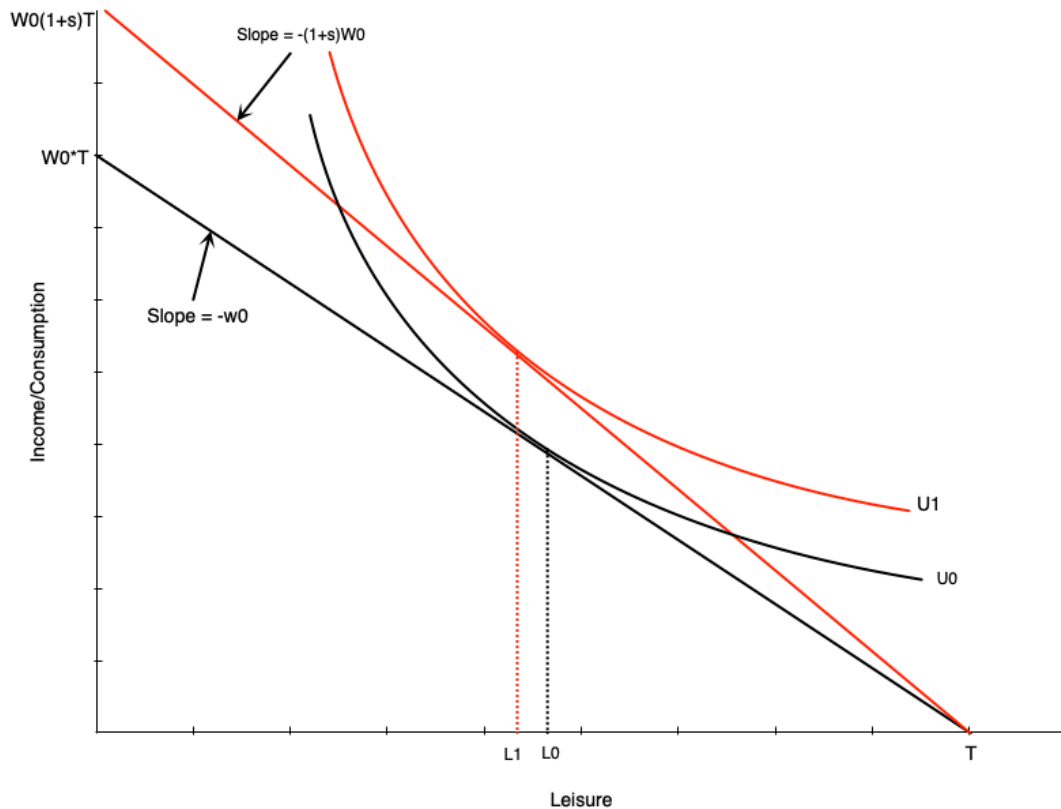
- Small decline in hours worked
  - Mainly mothers of newborns
  - Women in two-parent families
  - No significant decline for men

# Wage Subsidies: Theory

What is a wage subsidy?

- Payment to workers that increases their wage rate
- Example: \$5/hour subsidy increases \$15/hour wage to \$20/hour

# Wage Subsidies: Diagram



## Budget line pivots up with subsidy

- Graph shows person working more
- But effect is theoretically ambiguous
- Income and substitution effects work in opposite directions

# Wage Subsidies: Policy Implications

## Advantages:

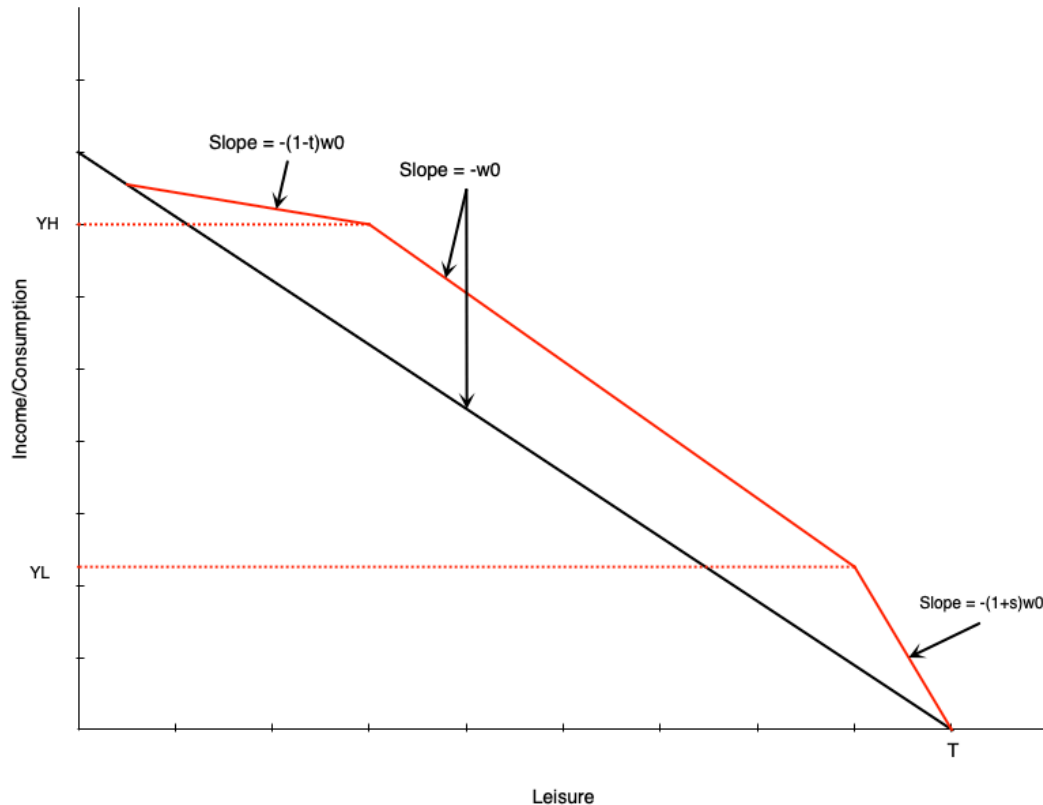
- Less negative work incentives than welfare or NIT
- No reduction in effective wage rate

# Tax Credits: Overview

Two types of tax credits:

1. **Non-refundable:** Can reduce tax owed to zero (but not below)
2. **Refundable:** Can reduce tax owed below zero (results in payment)

# Tax Credits: Budget Constraint



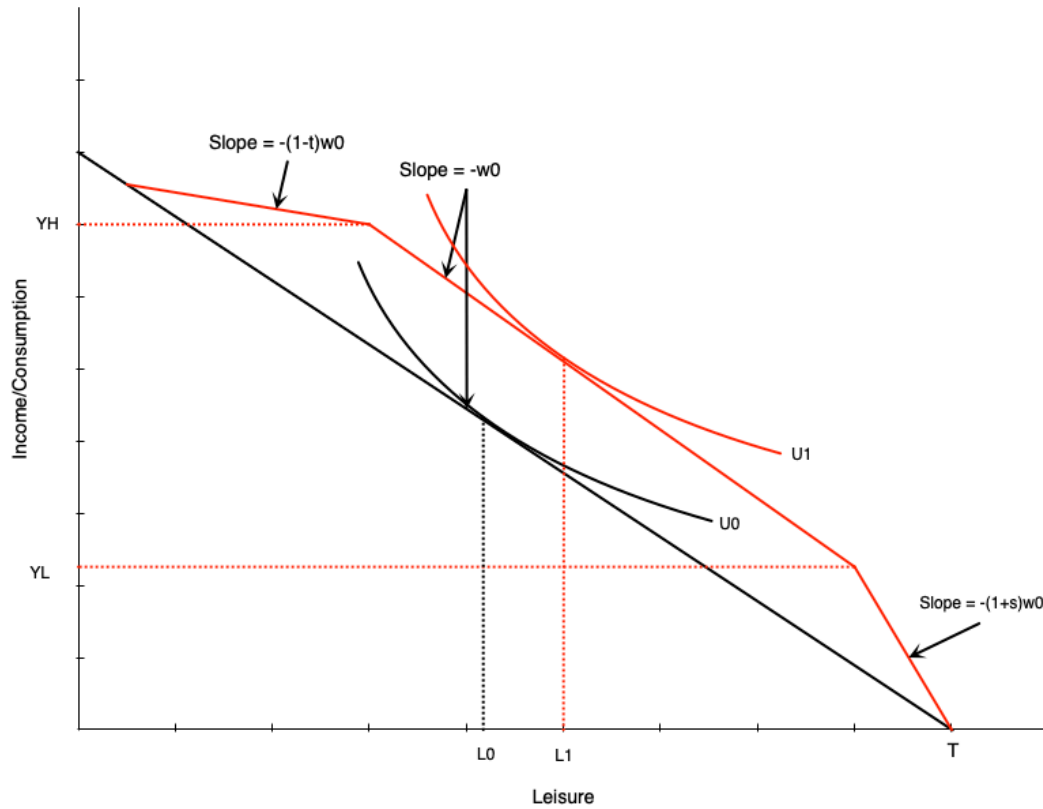
Three segments:

1. **Phase-in:** Subsidy proportional to earned income
2. **Plateau:** Maximum benefit level
3. **Phase-out:** Benefit reduced with earned income

Combines elements of:

- Wage subsidy (phase-in)
- Demogrant (plateau)
- NIT (phase-out)

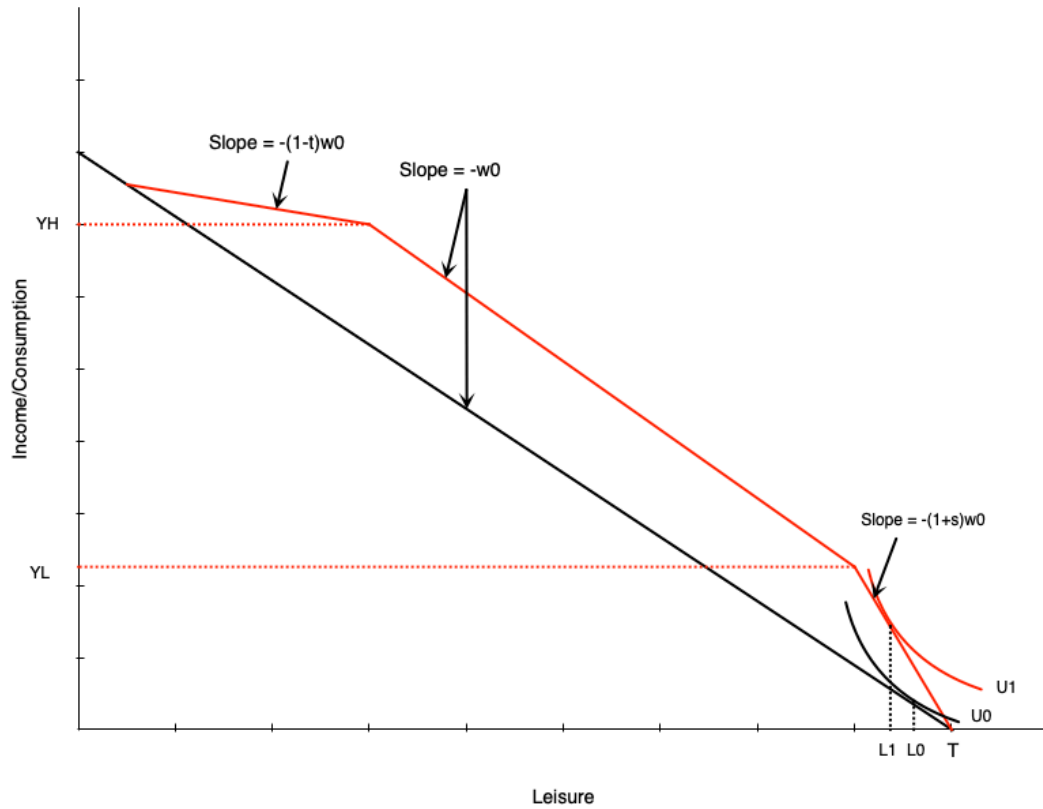
# Tax Credits: Effect in Plateau Range



Person initially in middle range:

- Tax credit gives maximum benefit
- Acts like **pure income effect**
- If leisure is normal good  $\rightarrow$  reduce work hours

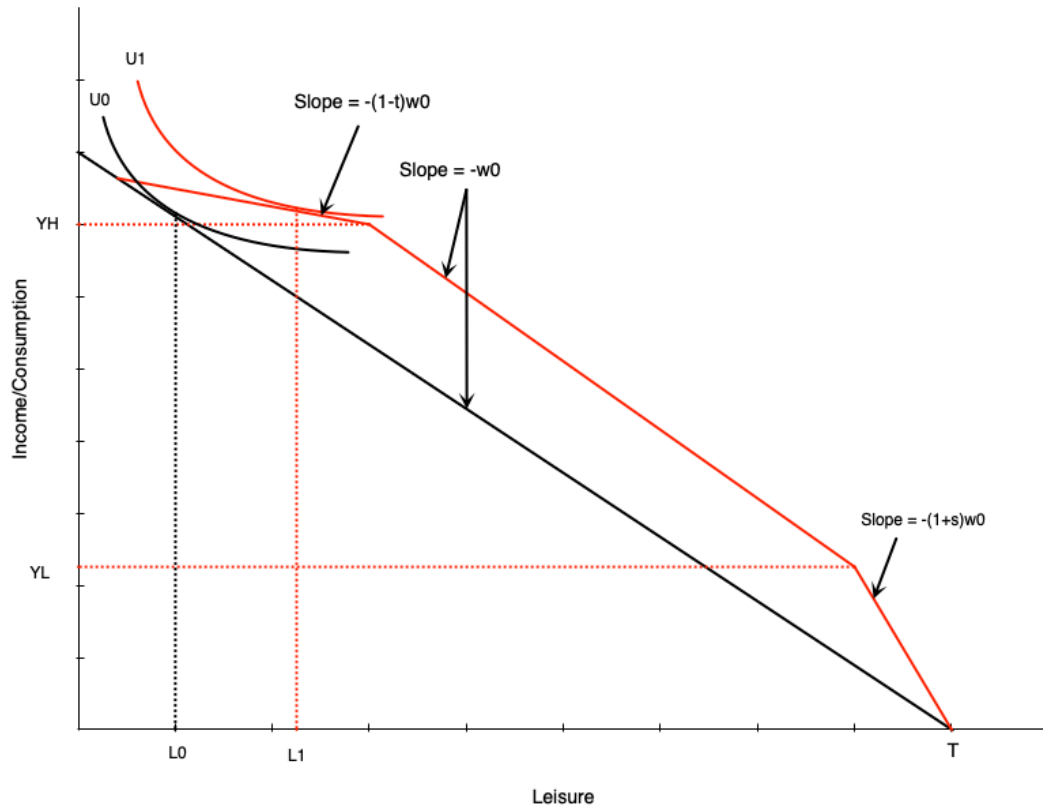
# Tax Credits: Effect in Phase-In Range



Person in phase-in range:

- Sees increase in effective wage
- Acts like **pure wage subsidy**
- Income and substitution effects in opposite directions
- **Effect on hours is ambiguous**

# Tax Credits: Effect in Phase-Out Range



Person in phase-out range:

- Similar to **negative income tax**
- Income effect  $\rightarrow$  reduce hours  $\downarrow$
- Substitution effect  $\rightarrow$  reduce hours  $\downarrow$
- **Effect on work is unambiguously negative**

# Unemployment Insurance

# Employment Insurance: Overview

## Purpose:

- Safety net for workers who lose their job
- Collect income while unemployed

# Employment Insurance: Benefits

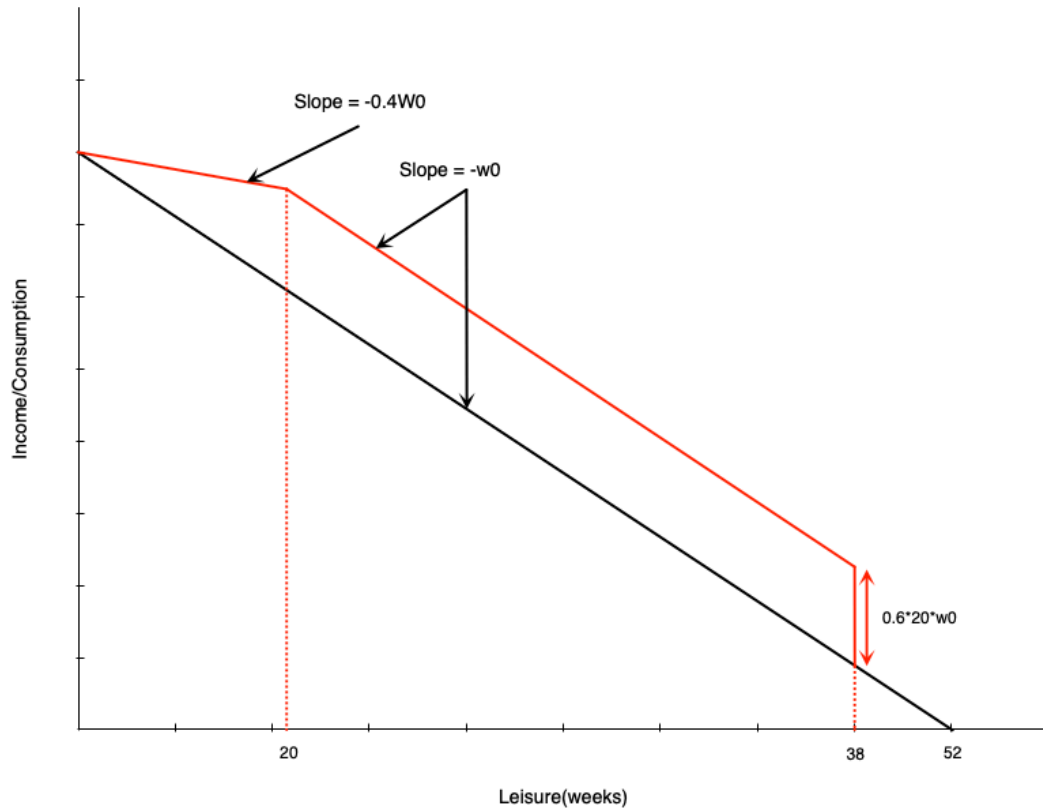
## Regular benefits:

- 55% of average insurable weekly earnings
- Up to a maximum amount
- Duration: **14 to 45 weeks**
  - Depends on unemployment rate and hours worked in last year

# El: Simplified Model Parameters

To analyze with income-leisure diagram:

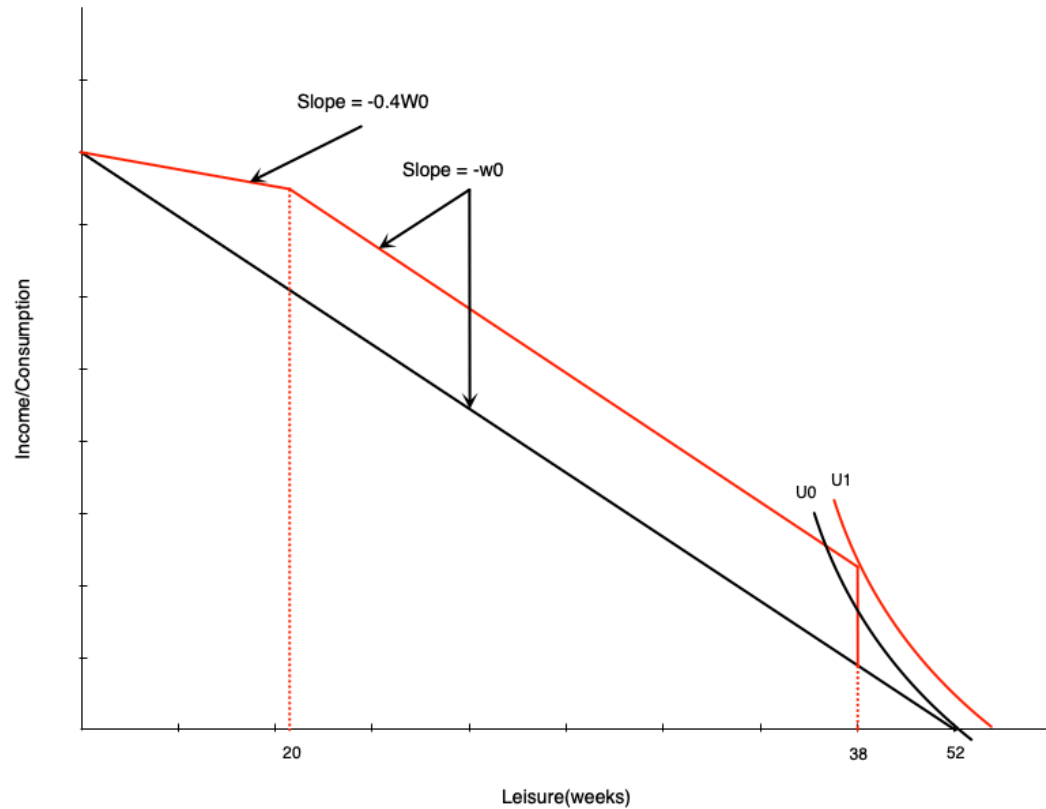
# El: Budget Constraint



Three segments:

1. **First 14 weeks:** Paid at usual wage  $w_0$
2. **Weeks 14-32 of work (20-38 weeks leisure):** Get 20 weeks of benefits
  - Benefit =  $0.6 \times 20 \times w_0$
3. **More than 32 weeks of work (<20 weeks leisure):** Benefits reduced week-for-week
  - Benefits paid at 60% of wage
  - Work paid at 100% of wage
  - **Keep 40% of wage** in this range

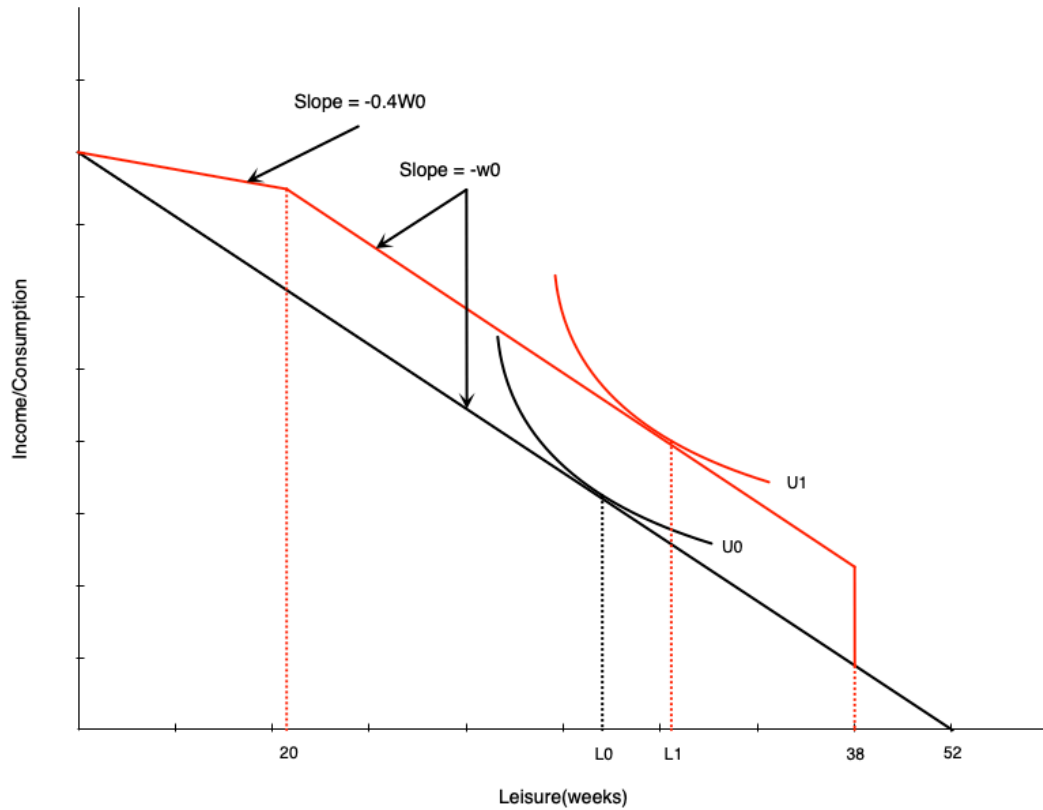
# EI: Effect on Non-Participants



A non-participant might start working:

- Large increase in non-labour income after 14 hours of work
- Induces labour force entry

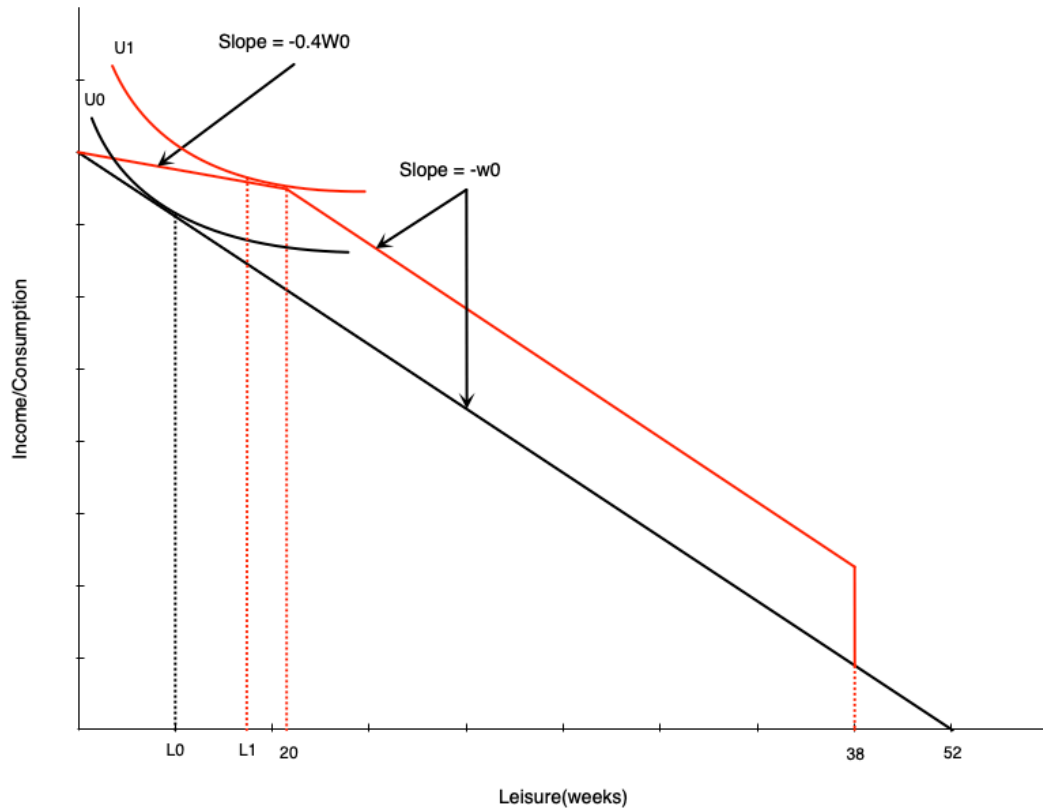
# EI: Effect in Middle Range



Person getting full benefits:

- Acts like **pure income effect**
- Will reduce work weeks

# EI: Effect on High-Hour Workers



In phase-out range:

- Acts like **negative income tax**
- Income effect  $\rightarrow$  reduce hours  $\downarrow$
- Substitution effect  $\rightarrow$  reduce hours  $\downarrow$
- **Effect unambiguously negative**

# Evidence: Maine vs. New Brunswick

Natural experiment:

# Evidence: Maine vs. New Brunswick Results

## Analysis findings:

### New Brunswick experienced:

- Higher labour force participation for marginally-attached workers
- Lower hours for those with stronger attachments

### Interpretation:

- Corresponds exactly to model predictions
- Highlights work disincentive effects of income maintenance programs

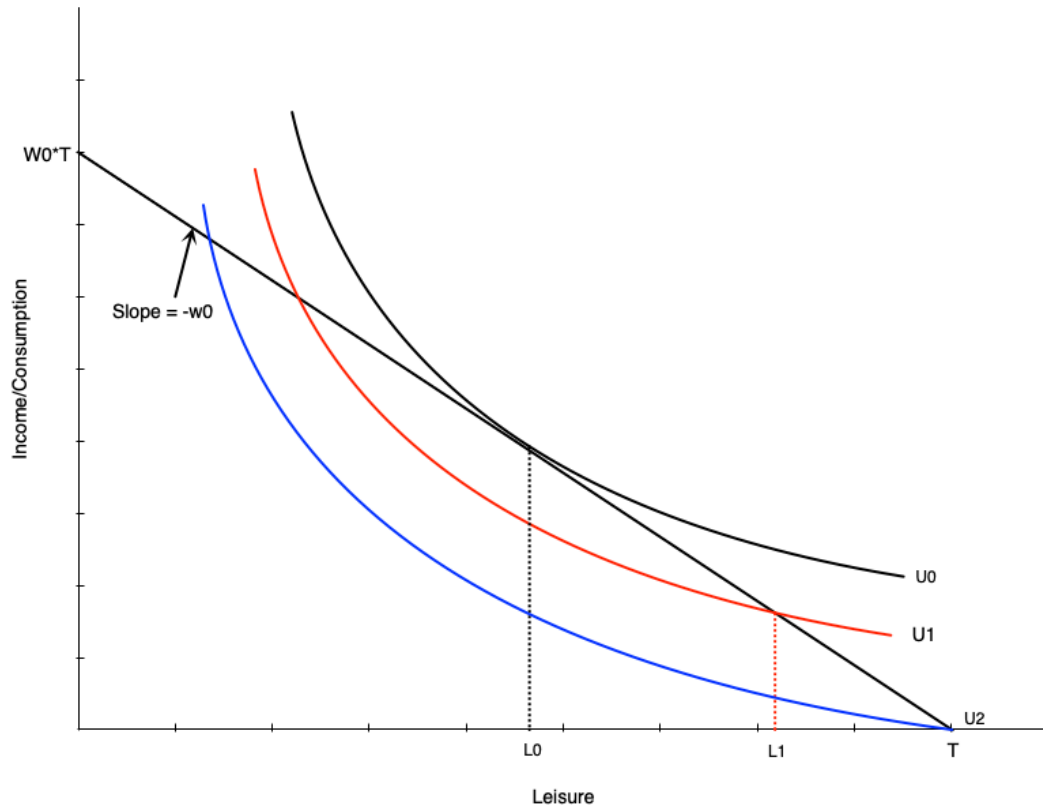
# Disability and Workers Compensation

# Introduction

What are these programs?

- Provide income support to injured/ill workers
- Come from public and private sources
- Premiums paid by employers and employees

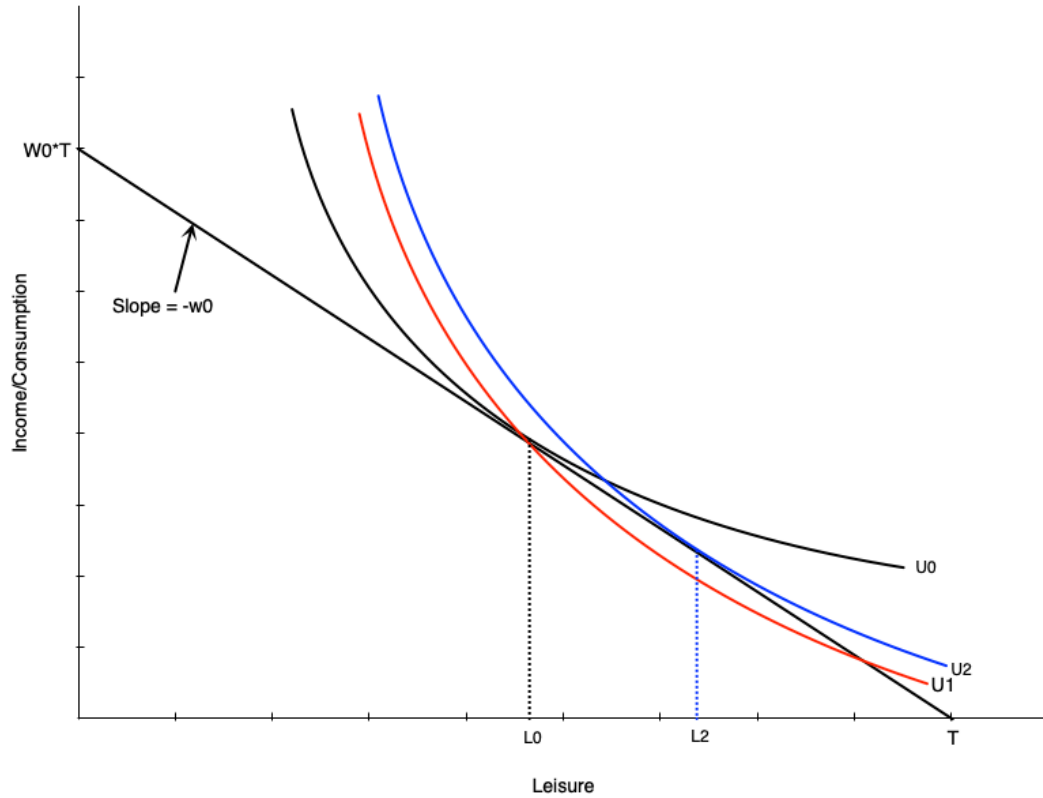
# Effect of Disability on Labour Supply



Three scenarios:

1. **Initially:** Choose  $L_0$  leisure, work  $H_0$  hours
2. **Injured but can work:** Choose  $L_1$  leisure, work  $H_1$  hours
  - Lower (non-optimal) indifference curve
3. **Completely disabled:** Choose  $T$  leisure, zero work
  - Even lower indifference curve

# Disability and Preferences



Disability may change preferences:

- Work becomes more difficult
- Preferences tilt toward leisure
- Indifference curve becomes steeper
- MRS increases at each point
- More willing to give up consumption for leisure

...

**Result:**

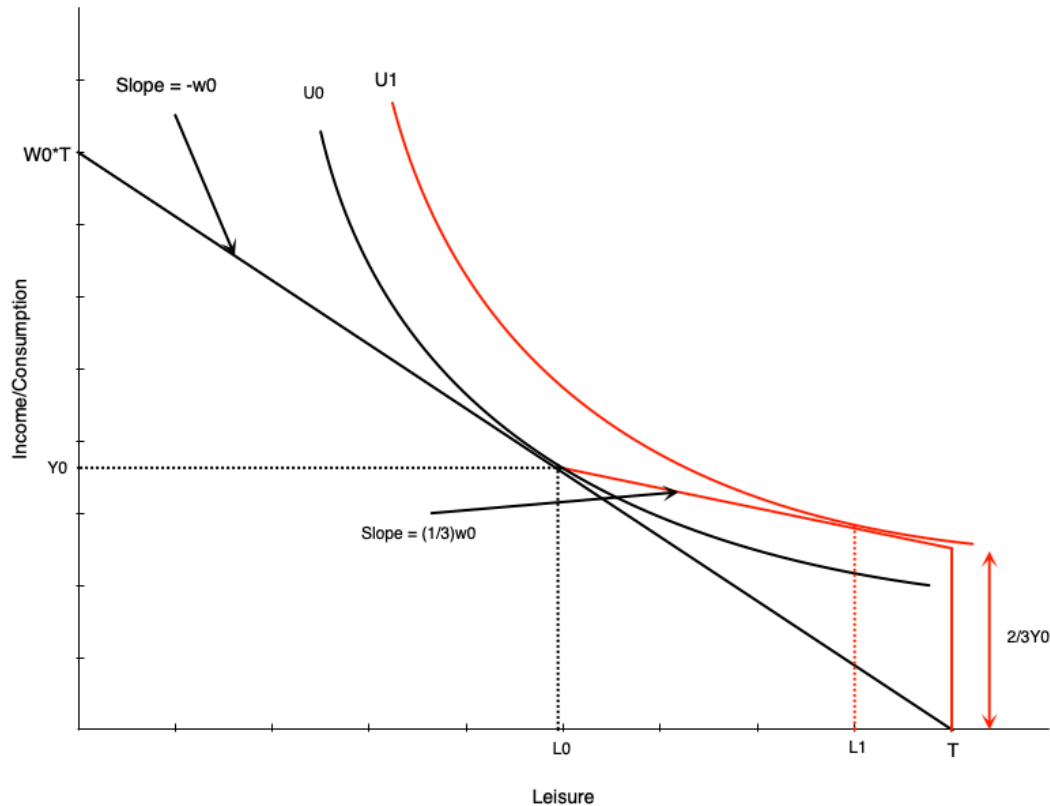
- Even if unconstrained, choose more leisure
- Person moves from  $L_0$  to  $L_2$

# Compensation Schemes

Typical disability insurance structure:

- Pays a fraction of previous earnings
- **Example (Ontario WSIB):**
  - 85% of net earnings (up to maximum)
  - Lasts until return to work or age 65

# Two-Thirds Compensation



Looks like negative income tax:

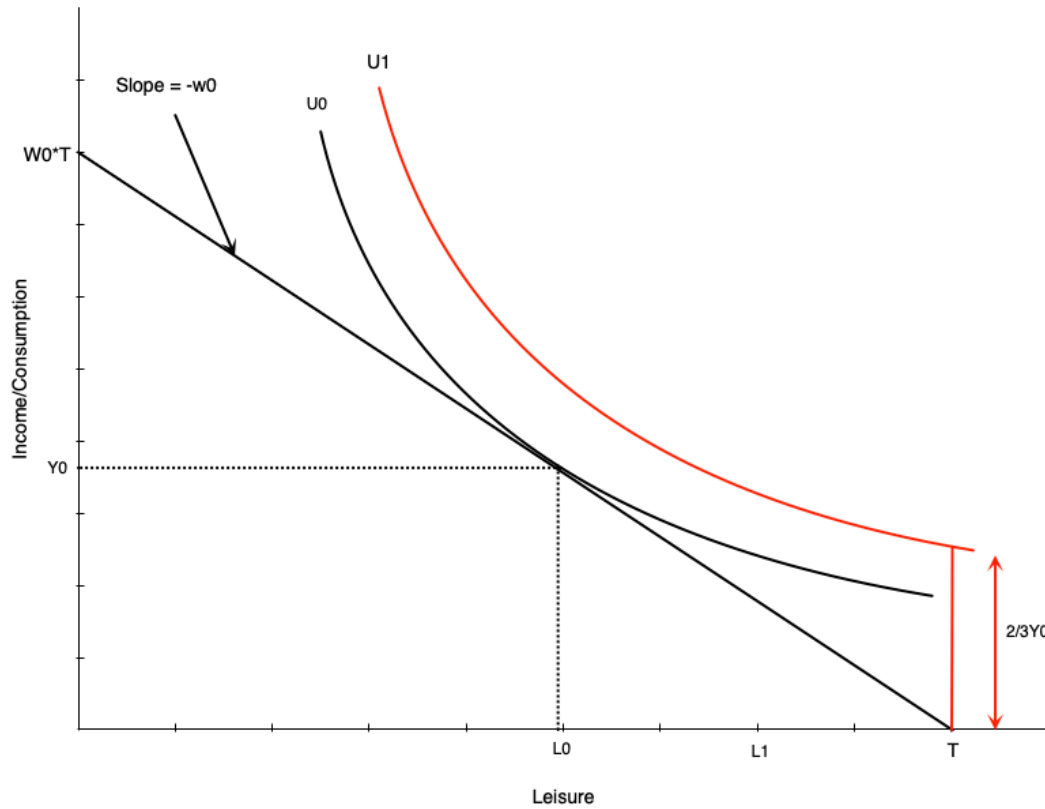
- At zero hours: get 2/3 of previous income
- For each hour worked:
  - Gain full wage
  - Lose 2/3 of wage in benefits
  - **Keep 1/3 of wage**

...

**For workers with flexible hours:**

- Effects match NIT
- Negative effect on hours worked

# Restricted Hours: Generous Compensation



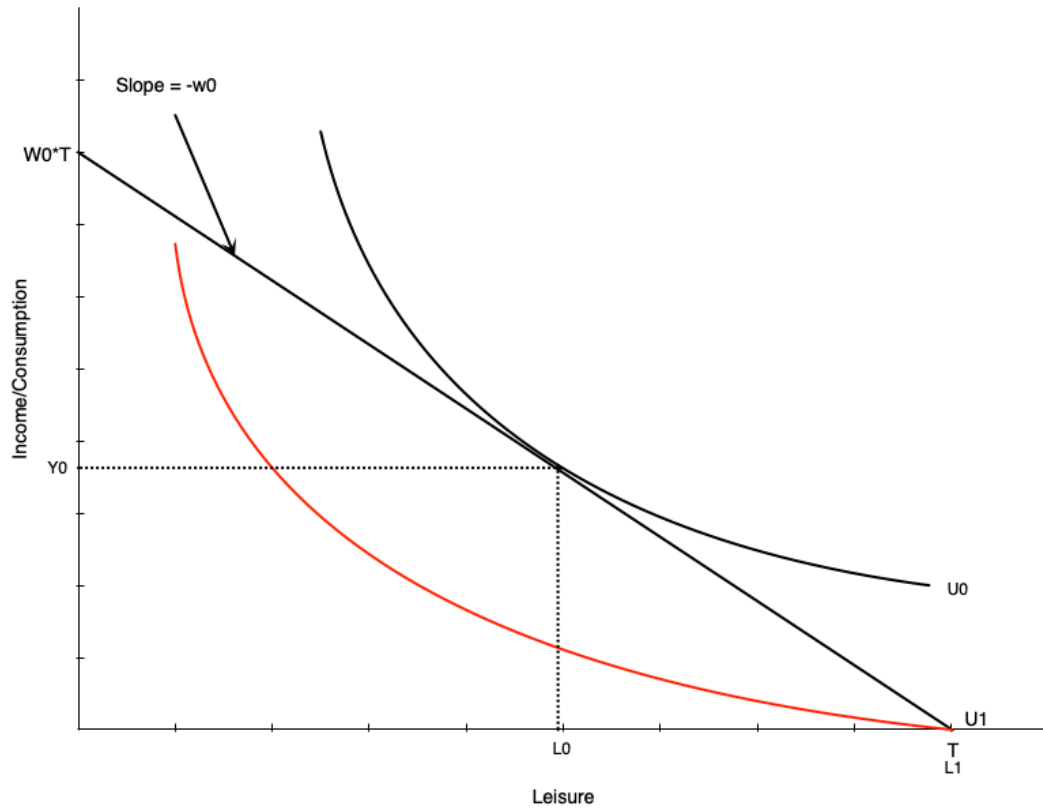
What if worker can only work  $H_0$  hours or zero?

- If scheme generous enough  $\rightarrow$  choose zero hours
- Utility at zero hours  $>$  utility at  $H_0$  hours

**Implication:**

Strong work disincentives of generous scheme

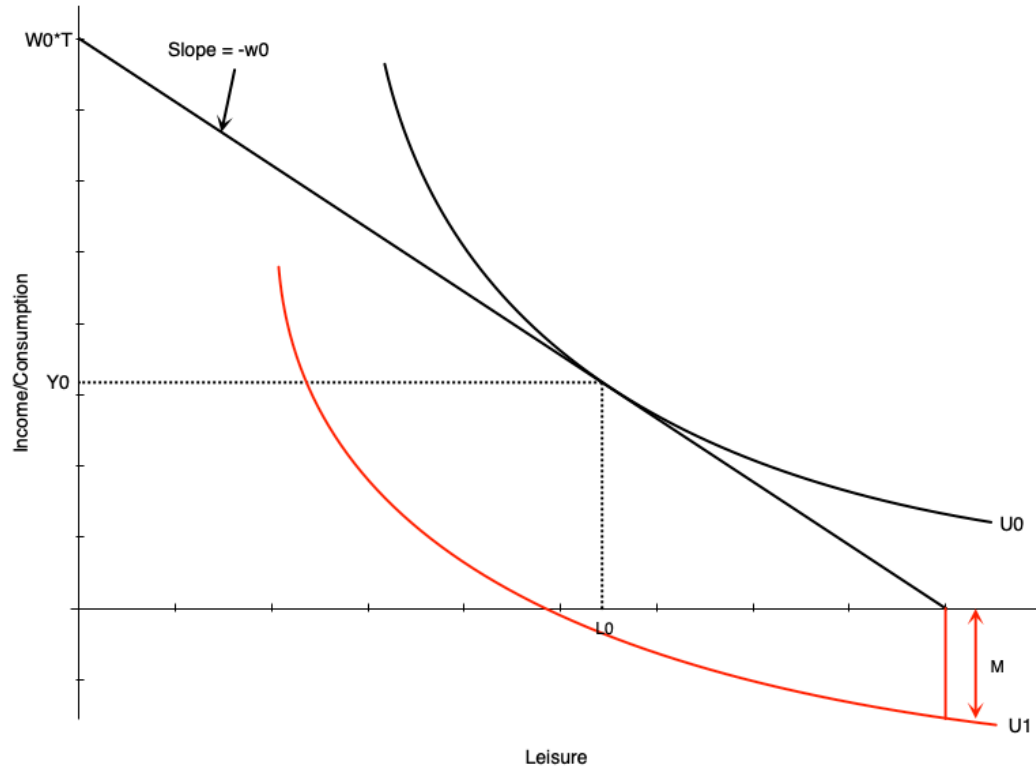
# No Compensation Case



Not providing compensation is problematic:

- Person who can work chooses  $H_0$  hours
- Person who cannot work gets lower utility
- End up substantially worse off

# Medical Expenses Case

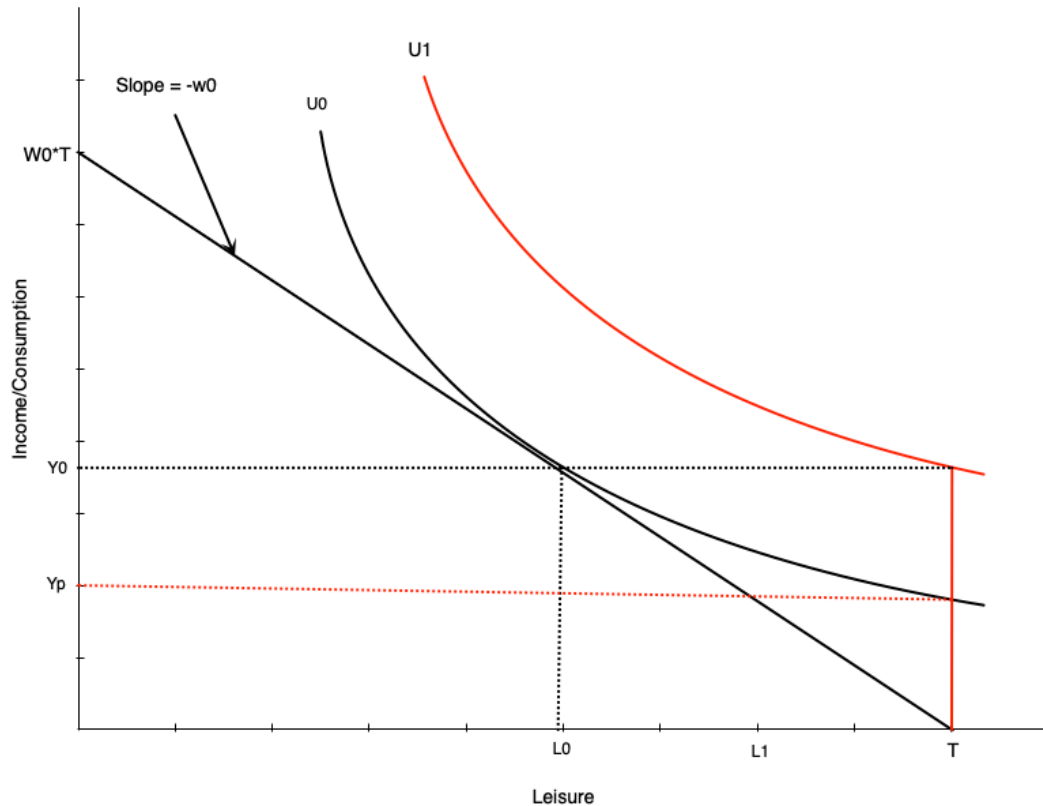


Medical expenses worsen situation:

- Act like reduction in non-labour income
- If large enough, non-labour income could go negative
- Occurs with no/very low compensation

This is an extreme scenario

# Optimal Compensation Level



How much to compensate non-workers?

Two options:

## 1. Full income replacement

- Higher indifference curve ( $U_1$ ) than working

## 2. Same satisfaction as working

- Same indifference curve as working
- Lower compensation ( $Y_p$ )

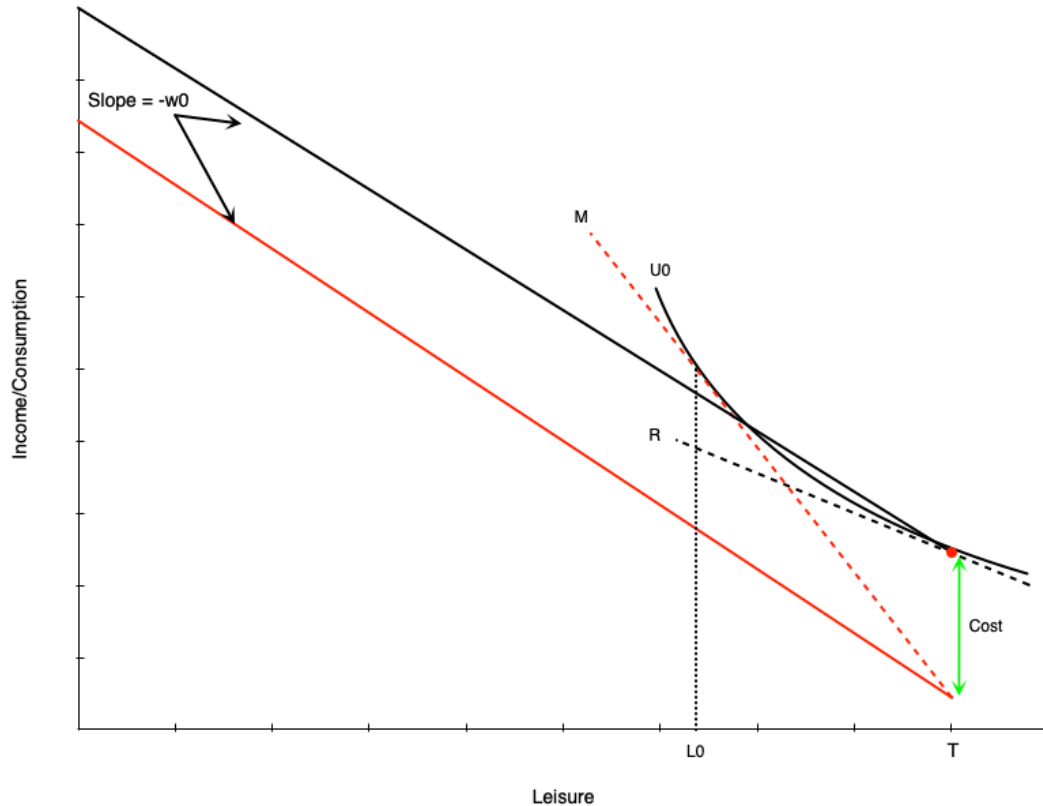
The optimal level is a policy question

# Child Care Subsidies

# Introduction

Canada's child care expansion:

# Child Care Costs: Non-Working Case



Budget line becomes discontinuous:

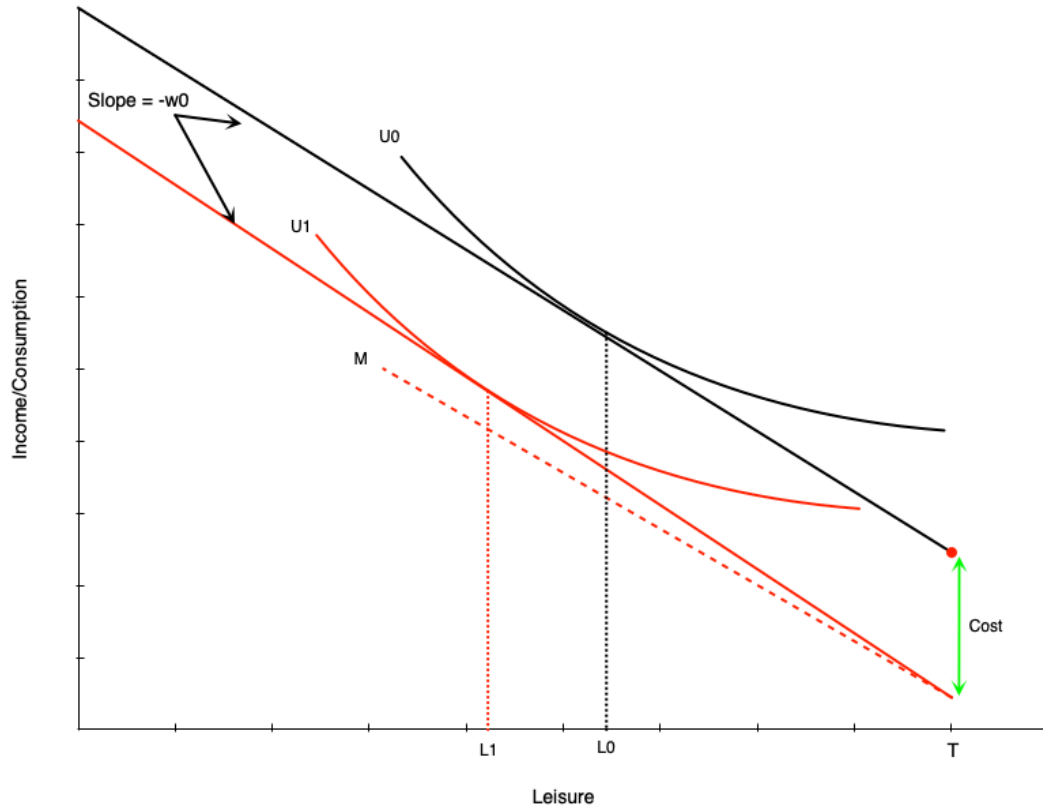
- At zero hours: red dot
- For any work: drops by child care costs
- Then straight line with slope = wage

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Effect:

- Child care costs **increase reservation wage**
- Need higher wage to induce work
- This person chooses not to work

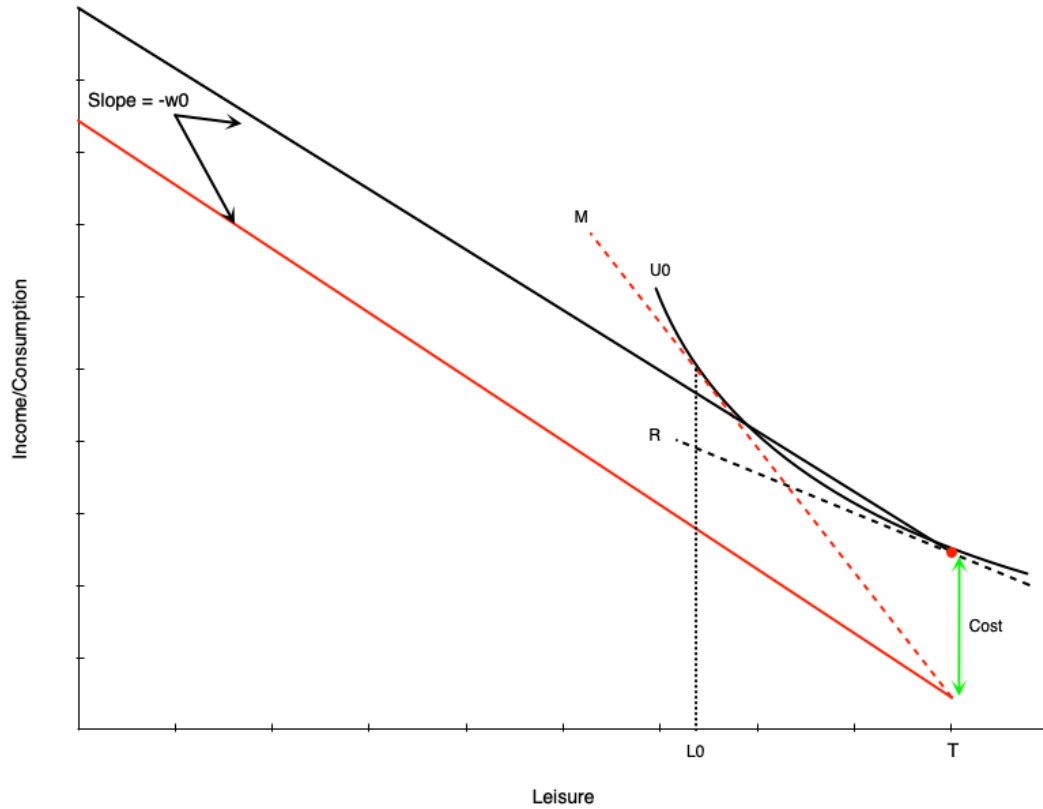
# Child Care Costs: Working Case



Person with reservation wage above wage + costs:

- Works positive hours
- Child care costs = **negative pure income effect**
- Choose less leisure, more work
- Must work more to offset income loss

# Child Care Costs: Labour Supply Gap



Creates gap in labour supply schedule:

- Person who works will work minimum of  $H_0 = T - L_0$  hours
- Not worth working less due to large child care costs

**Result:**

Labour supply schedule jumps from zero to  $H_0$  hours

# Child Care Subsidies: Effects

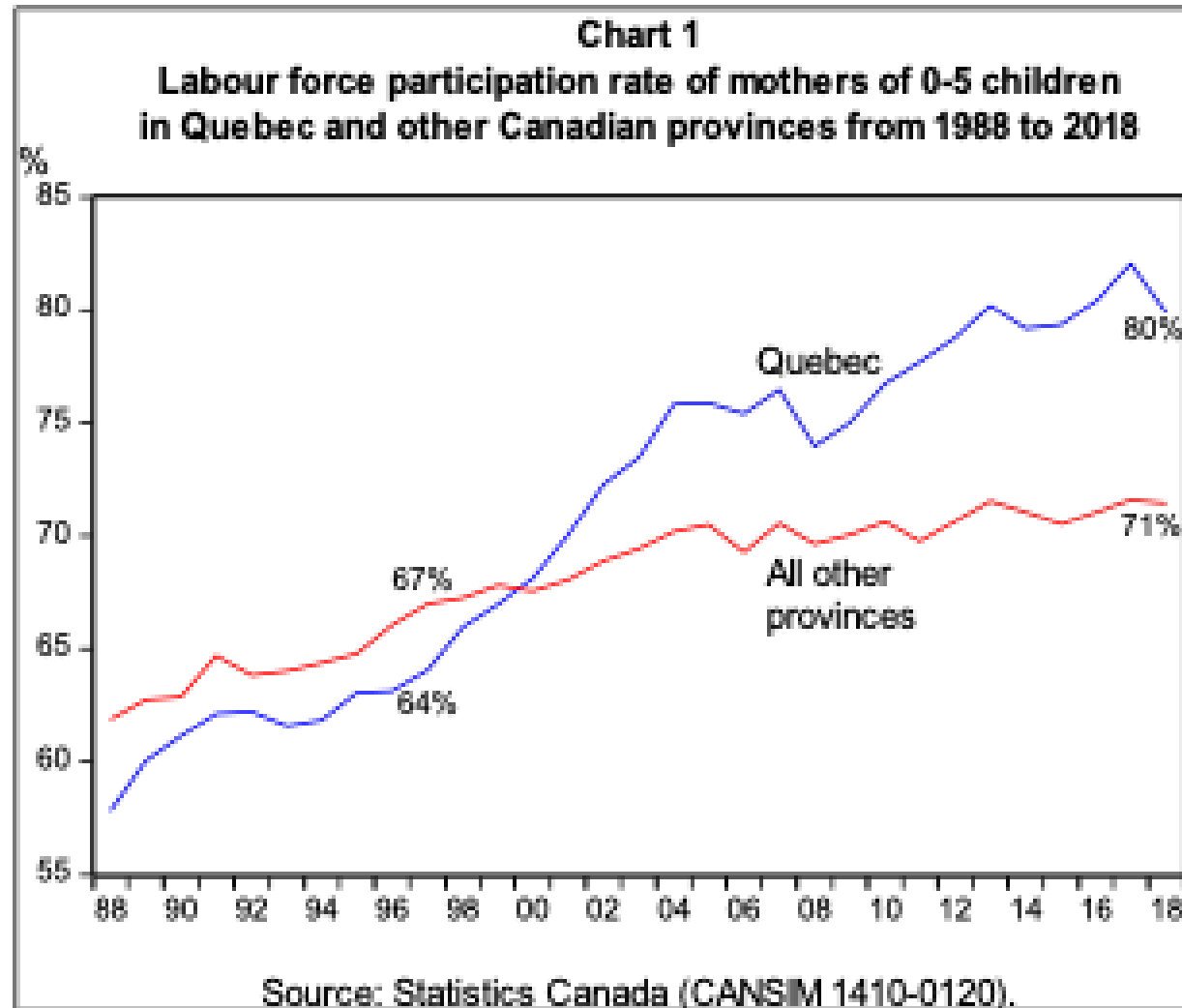
A full subsidy (costs  $\rightarrow$  zero) would:

# Evidence: Quebec's \$5/Day Program

## Program details:

- Implemented in 2000
- Currently sliding scale: \$8.25 to \$21.45

# Evidence: Visual Summary



Source: Fortin (2019)

# Summary

# Key Takeaways

Income maintenance programs: